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Mixed friction: Critical assessment of engineering load sharing equations based on the Lambda-parameter

Charlotte Spies^{a,b,*}, Yash Parab^{b,c}, Arshia Fatemi^b^a University of Freiburg, Department of Microsystems Engineering, Georges-Köhler-Allee 103, 79100, Freiburg, Germany^b Robert Bosch GmbH, Corporate Research and Advance Engineering, Robert-Bosch-Campus 1, 71272, Renningen, Germany^c University of Kaiserslautern-Landau (RPTU), Gottlieb-Daimler-Straße, 67663, Kaiserslautern, Germany

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ABSTRACT

In many engineering applications, such as gears or bearings, a load sharing concept between fluid friction and solid friction is typically used for the prediction mixed friction. The equations of sharing of load or shear stress are usually a function of the Lambda-parameter (λ -parameter) defined as the ratio between a theoretical film thickness and a single roughness metric. In the first part of this paper, we discuss from different perspectives why a single-metric load sharing method is not adequate for the differentiation of surface qualities concerning mixed lubrication and consequently prediction of friction.

In the second part of the paper, artificial roughness profiles were generated with an identical parameter set of power spectral density, and same roughness metrics of root-mean-square roughness S_q and average roughness S_a , but using three different height distributions (Gaussian, Bi-Gaussian and Weibull), and three different roughness orientations (isotropic, longitudinal and transverse). We show that surfaces with same roughness metrics, reveal very different load sharing and friction. Thus, the new load sharing concepts must entail additional roughness parameters to incorporate the effect of distribution and anisotropy. The comparison of the effect of roughness distribution and anisotropy has been performed with a numerical code for a thermal elastohydrodynamic lubrication for a point contact on a contact geometry of a conventional Mini Traction Machine. Finally some proposals for future load sharing concepts are given.

1. Introduction

The general trend in development of high energy efficient and high power density machine elements lead to system designs with lower viscosity and higher contact pressures. The former is mainly motivated by saving load-independent churning losses and load-dependent viscous losses. This caused that tribological contacts in machine elements such as gears and rolling element bearings work increasingly under mixed lubrication conditions [1,2]. It is especially more pronounced by the design of electrical drives, since the losses concerning power train constitute considerably higher percentage of overall losses compared to internal combustion engines [3]. Additionally, machine elements like piston rings and journal bearings operate in mixed friction conditions, highlighting the importance of this friction regime. Thus, practical engineering load sharing models are getting more essential for the prediction of mixed friction as well as the intensity of micro-stresses, electrical connection, heat transfer, noise, vibrations, etc.

The model complexities for load sharing in mixed lubrication range from highly sophisticated models including measured roughness deterministically coupled with elastohydrodynamic pressure [4,5], to

stochastic mixed friction analysis [6], to simple analytical engineering load sharing equations based on smooth surface film thickness and single roughness metrics as summarized by Taylor and Sherrington [7]. The former deterministic and stochastic models are not much used in engineering design, e.g., bearing and gear design software, because of their complexity and computational expenses as well as uncertainty of large influencing input parameter. The latter simple models, in contrast, are heavily used in all design software for gears and bearings, often without design engineers being aware of the limitations of these models.

The Lambda-parameter (λ -parameter) was originally used as a practical and useful engineering indicator for the evaluation of regimes of lubrication [8,9], and is generally defined as the ratio between some measure of film thickness over a metric describing the surface roughness. Lately, the machine element community has adopted it for a load sharing formulation and extended its application to predict mixed friction and surface fatigue [10]. It has been used also e.g., in ISO technical specification for the design of gears against micropitting [11] and also lifetime calculation of bearings (e.g., ISO 281:2007).

* Corresponding author at: University of Freiburg, Department of Microsystems Engineering, Georges-Köhler-Allee 103, 79100, Freiburg, Germany.
E-mail address: charlotte.spies@email.uni-freiburg.de (C. Spies).

This paper addresses the point that the widespread definitions of load sharing based on Lambda-parameters have some inherent physical inconsistencies. These inconsistencies cause various misconceptions resulted from assumptions and definitions which make the comparison of results of different laboratories and different calculation impossible even within same machine element community. In addition, using the Lambda-parameter in comparing and discriminating the frictional properties of different lubricated surfaces is only possible if the film thickness and surface roughness is exactly specified for the surfaces under investigation. So it renders the Lambda-parameter in many cases as not adequate, since it does not include any information about the nature of manufactured roughness textures (grinding, shaving, honing, polishing, etc.), state of surface (running-in), and the directionality of surface lay.

There are various critical assessment of the Lambda-parameter in the literature from different perspectives. Cann et al. [12] have shown the limitations of using the Lambda-parameter at low film thicknesses when compared to the surface roughness. They have shown that at higher roughness values compared to film thickness, local fluctuations of pressure on asperities cause elastic deformations of the surfaces. This leads to sensitivity to the direction of roughness, as well as to starvation conditions.

Zhu and Wang [13] consider the borderline between regimes of lubrication based on the Lambda-parameter as overly imprecise. They stress that these ranges do not adequately consider rough surface asperity interaction and the reduction of asperity height by surface deformation. With deterministic elastohydrodynamic lubrication (EHL) simulations and experiments they showed that when the Lambda-parameter is greater than 0.6–1.2, almost no asperity contact is found in neither experimental results nor numerical solutions (in contrast to classical ranges for asperity contact by $\lambda = 3$ or 4). Even for low Lambda values of about 0.05–0.1, a significant portion of the load (greater than 10–15%) could be supported by lubricant films. Recently, Zhang et al. [14] have shown that the Lambda-parameter might be applicable to the iso-viscous lubrication regime but not to the EHL regime and proposed instead a new parameter which includes the deformation of the roughness in highly concentrated piezo-viscous lubrication regime. For the evaluation of the regime of lubrication in gears and bearings, Hansen, Björling, and Larsson [15] proposed a new Lambda-parameter based on roughness parameters obtained from the Abbott-Firestone curve (due to its sensitivity to the running-in effect) and the radius of roughness peaks which includes the effect of micro-elastohydrodynamics of roughness in EHL. They did extensive studies how root-mean-square (RMS) roughness values are non-sensitive to the running-in effects. These studies have been continued by Prajapati, Hansen, and Björling [16,17]. Despite the clear limitations of using the Lambda-parameter with single roughness metrics, the usage of this method is very widespread in engineering because of its practical importance. In the next Section we investigate practical difficulties connected with this parameter, both as the indicator for the regime of lubrication, as well as the major parameter for load sharing.

2. State of the art of load sharing concept in engineering applications

This Section outlines the common inconsistencies inherent in application of the Lambda-parameter, highlighting the usage of different film thicknesses and roughness parameters which is practiced in engineering design.

2.1. Definition of film thickness

Film thickness in mixed lubrication system needs to be understood with special care, since in reality, in mixed lubrication, we can only talk about a film thickness distribution. By the presence of solid contact, the “real” minimum film thickness is actually zero. E.g., in point contacts

and at low amount of asperity contact sharing, a certain horseshoe shape can still be recognized and “theoretical” averaged minimum film on the constrictions at the sides are recognizable both in experiments and simulations [16]. However, at highly loaded systems with a prevalent asperity contact, the horseshoe effect almost vanishes and the “film thickness” can be defined as the separation between averaged lines of surface roughness as is the case by a mere contact mechanics analysis. We will show that in this case, the separation is rather defined by the asperities and the lubricant could have a very marginal effect [18].

2.1.1. Central, minimum or average film thickness

The film thickness equations in the machine design community are well established and have shown to be highly precise for central and minimum film thickness for different kinds of contacts, a.o., for point contacts [19], wide elliptical contacts [20], and slender elliptical contacts [21,22]. For friction prediction, i.e., for the determination of the Lambda-parameter, some publications used the central film thickness [10,23] while other publications used the minimal one, e.g., in [7,24]. Some publications do not give any clear indication whether minimum or central film thickness have been used [11,25]. It must be taken into consideration that the central film thickness can be up to 1.5 to 3 times higher than minimum film thickness [26] which makes a significant difference in the evaluation of the Lambda-parameter.

As stated, there is no unified definition in the literature about the usage of film thickness in the Lambda-parameter formulation in friction prediction. In the definition of the Lambda-parameter on the one hand it makes sense to use the minimum film thickness $h_{\min}$ to check whether the contact is in mixed friction or not. The complexity lies in the fact that while the major area in EHL contacts (line or elliptical contact) has a film thickness magnitude close to the central film thickness h_c , the solid contact load is carried rather by the area with the minimum films. This fact renders both parameter of h_c and $h_{\min}$ a significant choice for build up of a load sharing model. It is why in the community of deterministic mixed-EHL modeling, rather an average film thickness h_{avg} has been taken into consideration [4,6,13]. By prediction of friction, the above mentioned diversity of definitions, causes a serious misconception. This aspect is addressed in more detail in the evaluation of results. Another issue which needs to be addressed briefly here is that in calculation of film thickness, there is no standardized method to characterize the pressure-viscosity coefficient and there is a vast variance for this parameter based on different measurement methods and different evaluation methods [27].

2.1.2. Film thickness definition when the theoretical value is smaller than roughness

The idea to consider the surfaces as smooth for film thickness calculation and divide the film thickness by roughness metrics is a useful practice for designers regarding the evaluation of the surface quality in lubrication. However, using this ratio for load sharing might be very misleading especially at low speeds, or when using low viscosity lubricants. At very low speeds or with low viscosity lubricants, the theoretically calculated film thickness is considerably smaller than the roughness scalar values like RMS roughness R_q (profile), or S_q (areal). In this case, the separation of the surfaces is rather defined by roughness and contact mechanics and not by physics of EHL. Under these conditions, all load sharing equations are misleading. There are only few equations for load sharing in the literature which consider this aspect where the film thickness equation does not go down to zero [18,28]. The investigators used the mixed lubrication equation coupled with roughness deformation and asperity loads. They offer the only possibility to calculate load-sharing at small film thickness area by introducing additional dimensionless parameters of roughness and hardness into the film thickness evaluation by using a single roughness metrics S_q with a Gaussian height distribution assumption.

2.1.3. Surface roughness deformation in tribological contact

Due to higher pressure at asperities in tribological contacts, the asperities deform or flatten and shift the center-line of undeformed asperities, so an EHL contact could operate without any contact even at very small values of the Lambda-parameter [13]. This has been studied by Lubrecht and Venner [29] who showed that the longer wavelengths deform first and at lower pressure compared to short wavelengths. This adds another dimension of complexity which classical notion of the Lambda-parameter cannot capture [9]. Hansen, Björling and Larsson [15] have incorporated this aspect in the calculation of a modified Lambda-parameter Λ^* which includes the asperity peak radius in the equation (this parameter is highly scale-dependent). However, capturing in-contact roughness deformation adequately must generally be done with deterministic EHL simulations.

2.2. Definition of roughness

Besides the film thickness, a metric accounting for the surface roughness is needed for the determination of the Lambda-parameter. Similarly to the previous addressed inconsistencies, there are various approaches to regard the influence of roughness. In the following, the roughness as the denominator of the Lambda-parameter is discussed, as well as the uncertainty of roughness measurement and its impact on this parameter.

2.2.1. RMS roughness, average roughness or maximum height

There is no consensus in the definition of the scalar parameter as the denominator of the Lambda-parameter, neither for regime recognition, nor as load sharing and friction characterization. While the majority of the previous publications use RMS roughness R_q (profile) or S_q (areal) values as the denominator [7,30], there are gear standards which use average roughness R_a (profile) or S_a (areal) for prediction of gear efficiency and micropitting [31]. There are also publications with a formulation of Lambda-parameter based on the maximum height R_z of surfaces [32]. The variance is even larger since in some cases the scalar parameter of a single surface is used [10] and in other cases the average value (arithmetic or quadratic) of both contacting surfaces is considered [33]. Recently, the modified Lambda-parameter proposed by Hansen, Björling, and Larsson [15] includes additional roughness metrics, namely the asperity peak radius, and the reduced peak height. The scale dependency, particularly of the peak radius, poses a challenge when using this approach [34,35].

2.2.2. Roughness measurement method and filtering

In engineering practice, various methods are used to measure the scalar values of roughness, e.g., various optical or stylus contact methods. These methods yield a large variance of scalar parameters which impacts the denominator of the Lambda-parameter in a considerable way [36]. There is also a significant difference on whether a line profile or an areal spot has been measured. The variance is especially reinforced when considering the diversity of different optical objectives, measurement lengths, resolutions, stylus diameters, filtering parameters of roughness and waviness, etc. [36,37]. For example, considering stylus measurements, the probe size, i.e., tip radius and angle, affects the obtained scalar parameter of RMS roughness R_q [37,38]. When including other measurement techniques, e.g., optical methods and atomic force microscopy, the determined values of R_q may range over multiple length scales for the same surface, as highlighted recently by the Surface-Topography Challenge [36]. The uncertainty of roughness parameters is also discussed and highlighted by Seewig [39]. As long as the measurement method of scalar parameters of roughness is not unified or well characterized, the possibility to give a unique and universal equation for the relationship between friction and Lambda-parameter is by definition excluded. Among the vast literature on mixed friction in engineering applications, there are few works which address this limitation, e.g., Moyer and Bahney [40]. They proposed a new

methodology for a unified filtering of the surface roughness in order to have a more appropriate composite roughness for a broader range of bearing sizes and loadings in bearing applications.

For laboratory scale EHL studies of point contact, e.g., Mini Traction Machine, Bonaventure, Cayer-Barrio, and Mazuyer [41] have introduced a filtering methodology in order to consider the significant roughness length scales for the load sharing concept. The longer wavelengths will be flattened out and only the shorter wavelengths are taken into consideration [29]. Despite having the utmost significance, none of these filtering methods could establish themselves in the practice of friction evaluation in machine elements as well as many other recent publications in mixed friction.

2.3. Sharing concepts in mixed lubrication for assessment of friction

For the modeling of mixed friction various sharing concepts are used and are addressed in the following. For all, the total coefficient of friction (COF) μ_{total} is assumed to consist of the weighted sum of the COF in solid friction μ_{solid} and the COF from full-film lubrication μ_{fluid} , as

$$\mu_{\text{total}} = X \mu_{\text{solid}} + (1 - X) \mu_{\text{fluid}} \quad (1)$$

However, the approach to determine the weight factor X differs. Some authors use area sharing concepts [23,42,43], some use load sharing [44–48], and few propose shear stress sharing [49]. Zhu [30] points out that in the case of boundary lubrication, the real area of contact is not equal to the nominal area of contact. Thus, in the case of area sharing the weight factor defined as the ratio between real and nominal area of contact is not equal to 1 and the total COF would include influences of fluid friction. In the case of load sharing, which uses the ratio between the load supported by the asperities over the normal load, for this extreme case the weight factor X is reasonably defined. However, in the latter case, the real area of contact is not considered but could heavily influence the resulting stresses. While not widely used, shear stress sharing could be a promising concept to account for both aspects. Additionally, a general drawback of the load sharing approach for rough surfaces in EHL contacts is that the deformation of roughness due to occurring pressure is not taken into consideration [13,29]. It has been shown that the so called “in-contact” roughness is considerably deformed based on the amplitude, wavelength and operating conditions. Except for some cases like in [18,28] this fundamental fact has not been addressed in engineering approaches.

In the following, means for determining the weight factor used in Eq. (1) are briefly shown.

2.3.1. Area sharing models

When using area sharing, the weight factor X_a is defined as the ratio between real and nominal area of contact,

$$X_a = \frac{\text{Real area of contact}}{\text{Nominal area of contact}} \quad (2)$$

While the nominal area of contact is determined fairly easily, to determine the real area of contact, different approaches can be used. Greenwood and Williamson [44], and Bush, Gibson and Thomas [46] use multi-asperity models, where roughness is assumed to be on the same scale, e.g., same radius of asperities in the Greenwood and Williamson model [44]. In contrast, Archard [50] and Persson [51] use multiscale models to compute the real area of contact where roughness is assumed to exist on many length scales. For all these models, the real area of contact can be determined and then be taken into relation with the nominal area of contact to obtain the weight factor X_a . While area sharing is used for determining the weight factor [23,42,43], as mentioned, drawbacks arise especially when the Lambda-parameter is small [30].

Table 1
Contact load ratios according to different models.

Model	Weight factor/Contact load ratio
Coy [47]	$X_l = 1 - \lambda/3$ for $\lambda < 3$, $X = 0$ else
Greenwood and Williamson [44]	$X_l = \frac{F_{3/2}(c,d)}{F_{3/2}(0)}$
Greenwood and Tripp [45]	$X_l = \frac{F_{5/2}(c,d)}{F_{5/2}(0)}$
Bush, Gibson and Thomas [46]	$X_l = \operatorname{erfc}\left(\lambda/\sqrt{2}\right)$

Table 2
Weight factors according to different models.

Model	Weight factor/Friction force ratio
Olver and Spikes [54]	$X = (1 + \lambda)^{-m}$
Castro and Seabra [55]	$X = 1 - 0.84\lambda^{0.23}$ for $\lambda < 2.1$
Taylor and Sherrington [7]	$X = \frac{1}{(1 + \lambda^k)^a}$

2.3.2. Load sharing models

In the load sharing approach the weight factor X_l is defined as the ratio between the load supported by the asperities over the normal load,

$$X_l = \frac{\text{Load supported by asperities}}{\text{Normal load}}. \quad (3)$$

Different ways to determine this relation have been proposed, and some are given in Table 1. Here, $F_{3/2}(x)$ and $F_{5/2}(x)$ are statistical functions defined by Greenwood and Williamson [44] for which Jedynak gave analytical solutions for Gaussian surfaces [52], c is a factor to account for the roughness probability distribution, and erfc is the complementary error function. Note that for the determination of the weight factor with the Greenwood and Williamson approach, a Gaussian height distribution is assumed [44].

2.3.3. Shear stress sharing with numerical simulations

The load sharing based on load and area both have limitations in representing the true nature of the sharing of shear stress. However, even when the load is supported solely by the asperities in contact the real area of contact influences the resulting stresses which can vary heavily. For true modeling of mixed friction, a shear stress sharing approach combines the two concepts and is used for the determination of the total COF by few authors [49]. In this case, shear stress must be shared among asperity contact and hydrodynamic contact in an iterative manner in accordance with the load sharing and integrating based on the area sharing. This cannot be done with simple analytical equations, so an extensive numerical approach is needed to make any conclusion about how shear stress which leads to friction has been shared between asperities and the fluid film. For this, deterministic as well as stochastic methods can be used [13]. While the stochastic methods have some limitations in evaluation of contact at very low values of the Lambda-parameter, the deterministic models are quite time consuming to be solved. In this study, a combination of stochastic and deterministic method has been pursued in order to get the maximum insight from numerical simulations. It must be added that the shear stress shearing in numerical simulations is a mathematical abstraction of the reality of interfacial phenomena involved in mixed friction and there is still room for fundamental research on this topic.

2.3.4. Other sharing models

Besides sharing concepts based on contact mechanics, other approaches to determine the weight factor X have been presented. Examples are given in Table 2. Here, we want to highlight especially the model by Taylor and Sherrington [7]. The factors a and k are to be determined depending on the respective application [53].

Table 3
Parameters for the artificial surface generation using the MATLAB applet by Pérez-Ràfols and Almqvist [56,57].

Description	Symbol	Value	Unit
Shortest wavelength	λ_s	0.008	Normalized
Wavelength ratio	λ_1/λ_s	125	Normalized
Roll-off wavelength	λ_r	0.0625	Normalized
Height probability distribution (type)	$\phi(z)$	Gauss, Bi-Gauss and Weibull	
Hurst exponent (self-affine fractal)	H	0.8	–
Peklenik number	γ	See Table 5	
Bi-Gauss distribution parameter	c	8	–
Weibull distribution parameter	k	2	–
Grid size	$x; y$	255; 255	μm
Number of nodes	$M \times N$	256 $\times$ 256	–

Table 4
Definitions for the Gaussian, Bi-Gaussian and Weibull height distributions [56,59].

Distribution type	Height distribution $\phi(z)$
Gaussian	$\phi(z) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{z-m}{\sigma}\right)^2\right]$ with standard deviation σ and mean m
Bi-Gaussian	$\phi(z) = \frac{1}{\sqrt{2\pi}\sigma_1} \exp\left[-\frac{1}{2}\left(\frac{z-m_1}{\sigma_1}\right)^2\right] + \frac{1}{\sqrt{2\pi}\sigma_2} \exp\left[-\frac{1}{2}\left(\frac{z-m_2}{\sigma_2}\right)^2\right]$ with standard deviations $\sigma_{1,2}$ and means $m_{1,2}$
Weibull	$\phi(z) = \frac{k}{c} \left(\frac{z}{c}\right)^{k-1} \exp(-(z/c)^k)$ for $z \geq 0$ Pérez-Ràfols and Almqvist [56] use scale parameter $c = \sqrt{\frac{1}{\Gamma(1+\frac{1}{k}) - \Gamma^2(1+\frac{1}{k})}}$ with shape parameter $k > 0$ and Gamma-function Γ

3. Design of simulations with artificially generated roughness

Artificial rough surfaces were generated using the MATLAB applet by Pérez-Ràfols and Almqvist [56,57]. Surfaces having three distinct height distributions (Gaussian, Bi-Gaussian and Weibull), and three distinct roughness orientations (isotropic, longitudinal and transverse) were generated and the combination of those distributions and orientations sums up to nine total generated rough surfaces. The surfaces were resized in height to match the RMS roughness of $0.10\mu\text{m}$ and are shown in Fig. 1. The artificially generated surfaces are available at <https://contact.engineering/> [58] for usage of other researchers, and are linked in Appendix.

The input parameters for the MATLAB applet by Pérez-Ràfols and Almqvist [56,57] are shown in Table 3. All surfaces are generated using the multiscale roughness approach and are self-affine fractal (Hurst exponent of 0.8) with a roll-off region. In the following, the different height distributions and surface orientations, as well as the roughness characterization are presented.

3.1. Height distributions

Three different height distributions are used. The definitions of the probability distribution ϕ of the surface height z , or short height distribution, of the Gaussian, Bi-Gaussian and Weibull type are given in Table 4. Surfaces with a Gaussian height distribution are found in many theoretical models and represent surfaces before running-in. In contrast, a Bi-Gaussian or Weibull height distribution can be typically found in run-in surfaces. The average height distribution of the three roughness orientations is shown for each type of distribution respectively in Fig. 2.

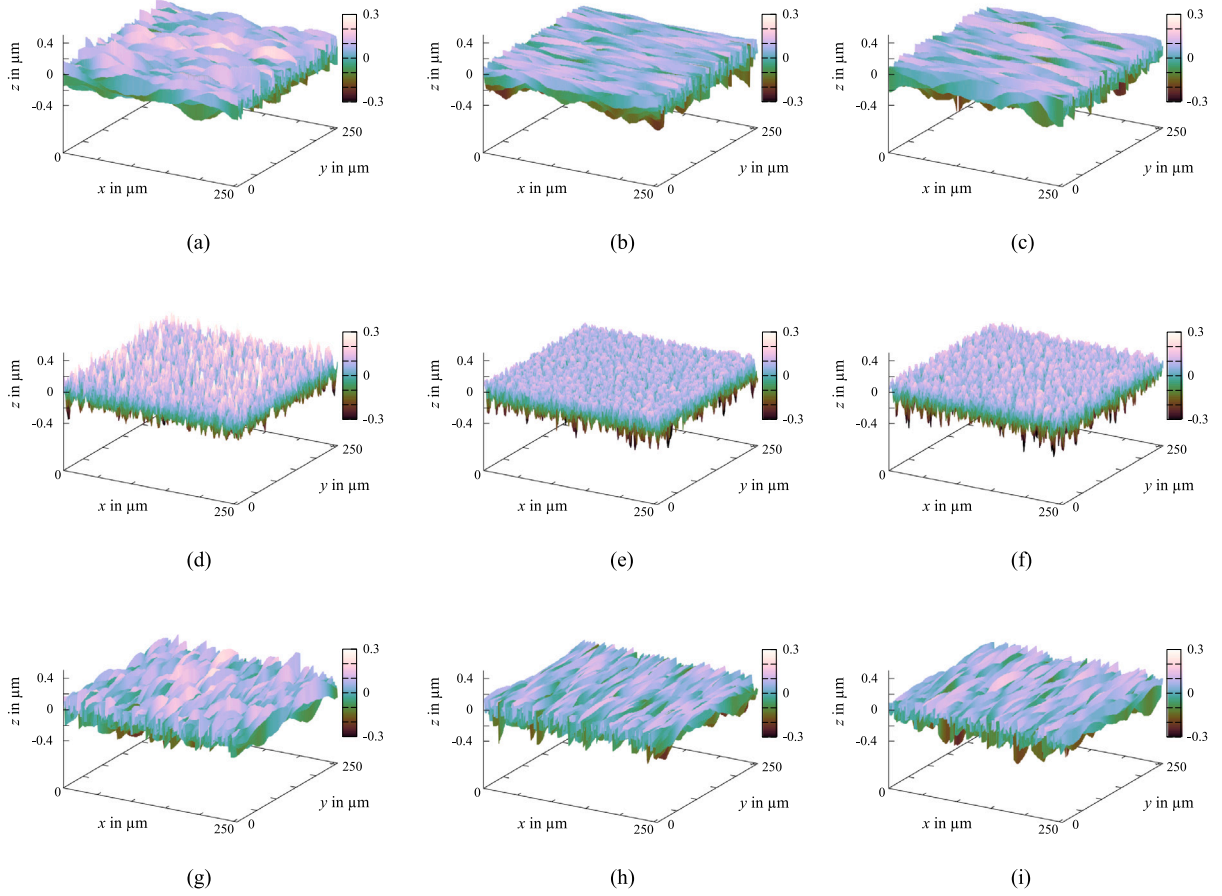


Fig. 1. Overview of the nine artificially generated surfaces with different height distributions and roughness orientations. Columns represent the height distributions, Gaussian, Bi-Gaussian, and Weibull (from left to right). Rows represent the roughness orientations, longitudinal, isotropic, and transverse (from top to bottom). (a) Gaussian height distribution, longitudinal roughness. (b) Bi-Gaussian height distribution, longitudinal roughness. (c) Weibull height distribution, longitudinal roughness. (d) Gaussian height distribution, isotropic roughness. (e) Bi-Gaussian height distribution, isotropic roughness. (f) Weibull height distribution, isotropic roughness. (g) Gaussian height distribution, transverse roughness. (h) Bi-Gaussian height distribution, transverse roughness. (i) Weibull height distribution, transverse roughness.

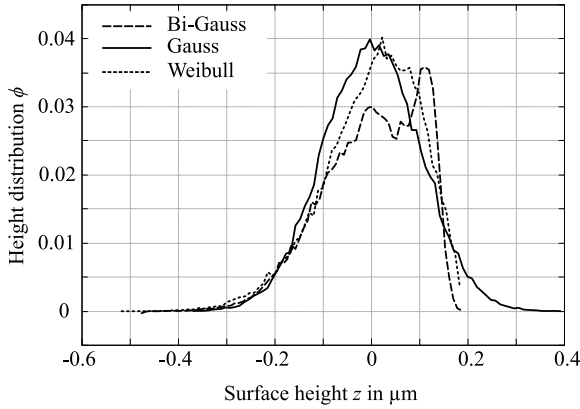


Fig. 2. Gaussian, Bi-Gaussian, and Weibull height distributions averaged over the three roughness orientations for the artificially generated surfaces.

3.2. Roughness orientation

In addition to the height distribution, the orientation of the surface roughness was varied. We generated isotropic, longitudinal and transverse surface roughness representing honed, shaved, and ground gear surfaces respectively. The roughness orientation is characterized with the Peklenik factor γ which is the ratio of the correlation lengths

in x - and y -direction, β_x and β_y . The characteristics of the roughness orientations are given in Table 5.

3.3. Roughness characterization

All nine artificially generated surfaces are designed to have the same RMS roughness S_q of $0.1 \mu\text{m}$. An overview of the classical roughness parameters, namely average roughness S_a , RMS roughness S_q , maximum height S_z , RMS gradient in x -direction R_{dqx} , and RMS gradient in y -direction R_{dgy} , as well as parameters from the Abbott-Firestone curve, particularly, reduced peak height S_{pk} , core height S_k , reduced pit depth S_{vk} , material ratio of hills S_{Mr1} , and material ratio of dales S_{Mr2} , are given in Table 6. Their definitions can be found in ISO 21920 [60]. Especially the average and RMS roughness S_a and S_q must be highlighted. Even though the surfaces differ in height distribution and roughness orientation, these values are constant. However, it can be clearly seen that the height distribution heavily influences the parameters of the Abbott-Firestone curve. In contrast, the RMS gradients R_{dqx} and R_{dgy} change based on the roughness orientation.

4. Effect of roughness distribution and orientation on flow factor simulation

In order to consider both, the effect of load and area sharing, as well as effect of deformation of roughness caused by pressure in the contact, flow factors and compliance-pressure curves (asperity contact

Table 5Parameters of roughness orientation and correlation lengths (corr. length) including Peklenik factor γ of the artificially generated surfaces.

Roughness orientation	Corr. length β_x (μm)	Corr. length β_y (μm)	Peklenik factor $\gamma = \beta_x/\beta_y$ (–)	Orientation representation
Longitudinal	40.18	0.64	63	Shaved (conventional) gear surfaces
Isotropic	2.60	2.65	1	Honed (external) gear surface
Transverse	0.64	40.18	0.02	Ground (indexing) gear surfaces

Table 6

Amplitude and hybrid roughness metrics, and parameters from Abbott-Firestone curve for the artificially generated surfaces.

Surface description	Amplitude parameters			Hybrid parameters		Parameters from Abbott-Firestone curve				
	S_a (μm)	S_q (μm)	S_z (μm)	R_{dqs} (–)	R_{dgy} (–)	S_{pk} (μm)	S_k (μm)	S_{vk} (μm)	S_{Mr1} (%)	S_{Mr2} (%)
Gauss, isotropic	0.080	0.10	0.834	0.06	0.06	0.089	0.244	0.089	9.7	90.0
Weibull, isotropic	0.080	0.10	0.749	0.06	0.06	0.038	0.244	0.119	5.9	87.3
Bi-Gauss, isotropic	0.082	0.10	0.645	0.06	0.06	0.013	0.271	0.105	0.9	89.2
Gauss, transverse	0.080	0.10	0.772	0.141	0.003	0.078	0.210	0.080	9.6	89.7
Weibull, transverse	0.080	0.10	0.754	0.140	0.003	0.041	0.212	0.103	6.1	87.4
Bi-Gauss, transverse	0.082	0.10	0.670	0.140	0.003	0.025	0.224	0.093	4.9	88.6
Gauss, longitudinal	0.080	0.10	0.772	0.003	0.141	0.078	0.210	0.080	9.5	89.7
Weibull, longitudinal	0.080	0.10	0.754	0.003	0.140	0.041	0.211	0.103	6.2	87.3
Bi-Gauss, longitudinal	0.082	0.10	0.670	0.003	0.141	0.024	0.224	0.093	4.9	88.6

curves) have been derived as correction factors in an average Reynolds equation [61]. The flow factors, were determined using the average flow model by Patir and Cheng [61] and were calculated deterministically based on the work of Lagemann [62]. The contact pressure curve and real contact area were calculated via boundary element method [63,64]. The definitions and equations will not be shown in detail here. The reader is referred to [61,62].

4.1. Pressure and shear flow factors

For the nine artificially generated surfaces, the pressure flow factors in x - and y -direction, Φ_x^P and Φ_y^P , are shown in Fig. 3. They are defined as the ratio of pressure-driven flow rate between rough surfaces to the pressure-driven flow rate between ideally smooth surfaces. In contrast, the shear flow factor describes the additional flow transport due to the surface roughness. The shear flow factors for both directions, Φ_x^S and Φ_y^S , are presented in Fig. 4. All flow factors are shown in dependence of the deformed gap size h_d which is the distance between center-lines of roughness profiles after deformation of roughness, due to the effect of micro-elastohydrodynamics (see [61,62,65]). A high deformed gap size h_d indicates no effect of roughness, while a small h_d represents impeding the flow through roughness. As expected, the pressure flow factors approximate 1 for high deformed gap sizes, while the shear flow factors asymptotically approach zero. Naturally, the pressure and shear flow factor for the isotropic surfaces does not depend on the direction, so that $\Phi_x^P(h_d) = \Phi_y^P(h_d)$ and $\Phi_x^S(h_d) = \Phi_y^S(h_d)$. However, the anisotropy can be clearly seen when comparing the flow factors in each direction in the case of the longitudinal and transverse roughness orientation.

The pressure flow factor describing the pressure-driven flow rate, see Fig. 3, shows a major influence by the surface anisotropy while the distribution of the surface height has almost no effect on the impediment of flow. However, the longitudinal roughness increases the flow rate in the x -direction whereas the transversal and isotropic one rather impede the flow hence yield to comparatively higher actual film thicknesses. For the pressure flow factor in y -direction the opposite behavior is seen. In the case of shear flow factors, see Fig. 4, the transverse roughness is the impeding one regarding the x -direction. Shear flow factors can be interpreted as a change in gap height caused by local displacement flows at the individual roughnesses. Since the influence of the local displacement flows increases with decreasing lubrication gap height, the shear flow factor also increases accordingly.

4.2. Asperity pressure and contact area ratio

The asperity contact curves p_{asperity} is shown in Fig. 5 in dependence of deformed gap height h_d for all nine generated surfaces. The height distribution influences the obtained asperity pressure, especially when comparing Bi-Gaussian or Weibull distributions with the Gaussian one. In comparison, the contact area ratio, i.e., the ratio between real over nominal area of contact, is given in Fig. 6. It can be observed that there is a clear differentiation in the onset of asperity contact by approaching the surfaces in both asperity load sharing and contact area ratio. Referring back to Eq. (1) and the inconsistency of the definition of the weight factor, the differences between load and area sharing are highlighted. Especially for small deformed gap heights h_d the differences between these two approaches can be seen in Figs. 5 and 6. While the height distribution influences the asperity pressure, Fig. 5, especially when comparing Bi-Gaussian or Weibull distributions with the Gaussian one, the contact area ratio does not show this clear behavior, Fig. 6. In contrast, for asperity load sharing pressure curves, the distribution plays a major role, and the Weibull and Bi-Gaussian surfaces show a very similar pattern. For area sharing, again the contrast between effect of distribution and roughness orientation is not very clear but generally the effect of distribution is more pronounced.

5. Mixed EHL simulation of surfaces with same power spectral density and different distributions and orientations on the geometry of mini traction machine

The mixed EHL simulations for point contacts were performed with a multilevel EHL solver. The simulations are performed in a contact geometry analogous to Mini Traction Machine (MTM) tests so that the results can be related to experiments. We use the geometry of ball and disk with a given slide-to-roll ratio of 5% to minimize the effect of temperature thinning of the lubricant. The parameters of the virtual MTM tests are given in Table 7.

Detailed descriptions of the working principle of the EHL solver can be found in [64,66], which are based on methodology presented in [63]. The modeling of roughness in the Reynolds equation is performed with stochastic surface roughness modeling (average flow model) and the flow factors (as shown in the previous Section) by the method presented by Lagemann [62], which is also used to deterministically acquire contact pressure curves.

The simulations are conducted under the assumptions of steady state, and fully-flooded contact conditions, i.e., no starvation. The ref-

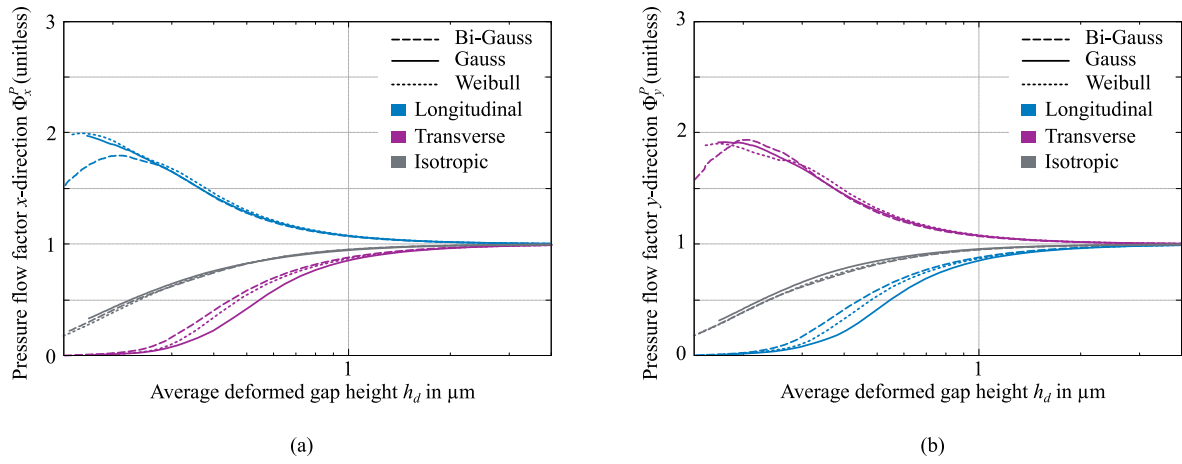


Fig. 3. Pressure flow factor Φ_x^P and Φ_y^P of the artificially generated surfaces as a function of deformed gap size h_d . (a) In x-direction. (b) In y-direction. The different height distributions (Bi-Gauss, Gauss, Weibull) are distinguished by different line types, and the roughness orientations (longitudinal, transverse, isotropic) are shown by the line color, resulting in the nine artificially generated surfaces.

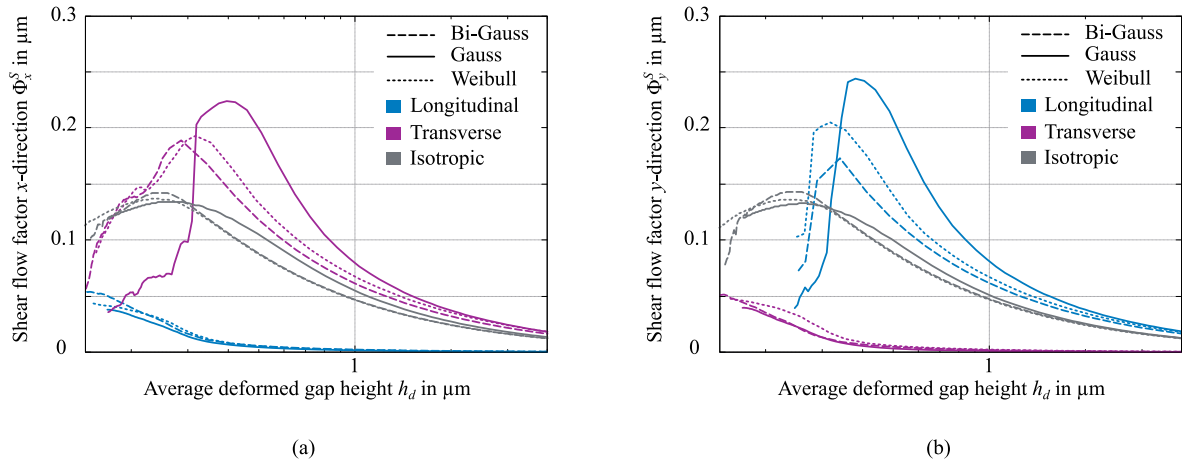


Fig. 4. Shear flow factor Φ_x^S and Φ_y^S of the artificially generated surfaces as a function of deformed gap size h_d . (a) In x-direction. (b) In y-direction.

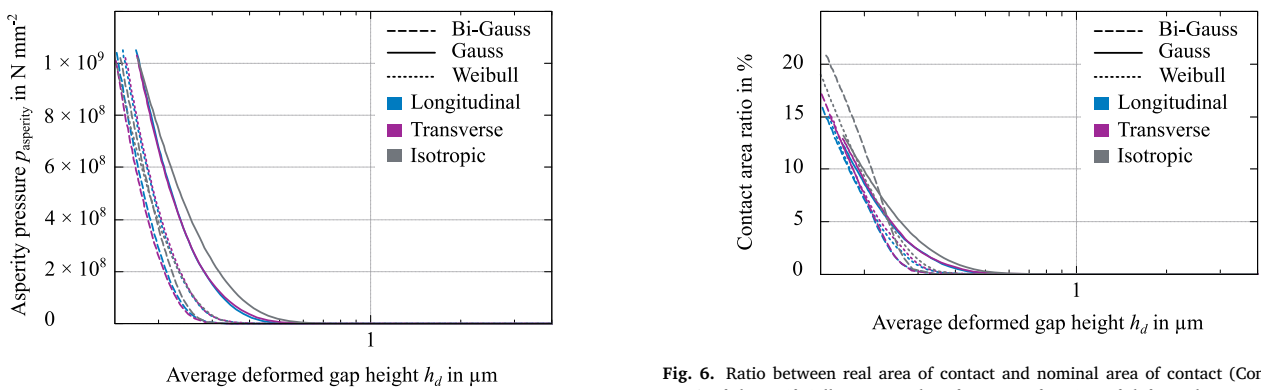


Fig. 5. Asperity contact pressure (compliance-pressure curves) p_{asperity} of the artificially generated surfaces as a function of deformed gap size h_d .

Fig. 6. Ratio between real area of contact and nominal area of contact (Contact area ratio) of the artificially generated surfaces as a function of deformed gap size h_d .

erence lubricant Tris(2-ethylhexyl) trimellitate (TOTM) is used and its viscosity–pressure relation is modeled based on the improved Yasutomi equation [27], whose parameters are given in Table 8. To describe the shear thinning, an Eyring model is considered. Table 9 shows our experimentally approximated rheological parameters obtained via characterization method.

5.1. Evaluation of EHL film thickness by presence of roughness anisotropy and distributions

The major causal argumentation in evaluation of friction can be found in the study of film thickness. The central film thickness obtained from the simulations is shown in Fig. 7a. In addition, Fig. 7b shows the minimum film thickness. The values are averaged based on the average flow model (the deterministic minimum film thickness by asperity

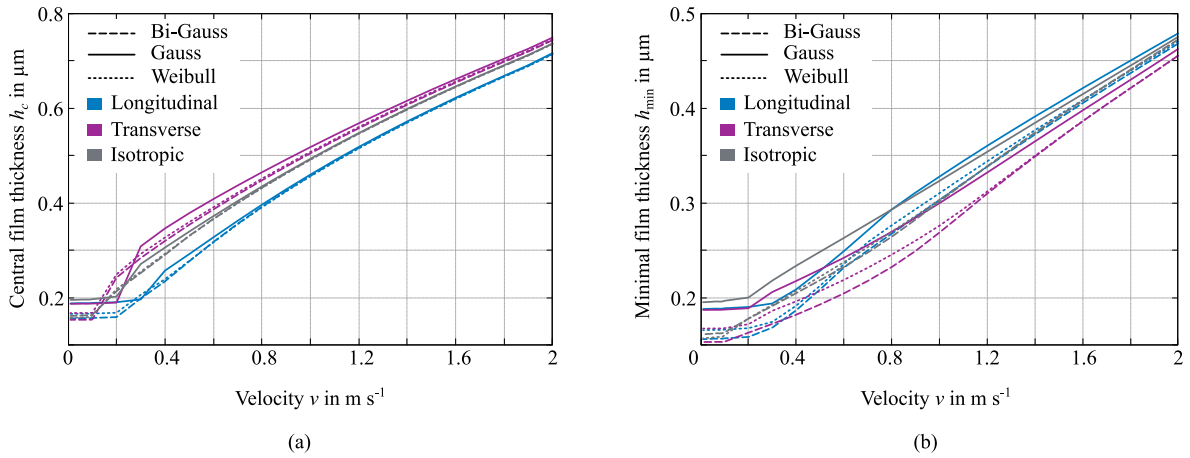


Fig. 7. Film thickness in dependence of velocity v . (a) Central film thickness h_c . (b) Minimal film thickness h_{min} .

Table 7

Parameters of the virtual MTM model.

Parameter	Value
Normal force	23 N
Young's modulus	210 GPa
Poisson's ratio	0.29
Entrainment speed	0.0–2.0 m/s
Ball diameter	19.50 mm
Temperature	40–100 °C
Slide-to-roll ratio	5%

contact will be zero). The fundamental insight that can be observed is that the longitudinal roughness gives the lower central film thickness compared to the isotropic and transverse one which is in accordance with the major published literature on numerical and experimental studies on point contact [69–71]. However, the simulation data from the minimum film thickness, see Fig. 7b, indicates that the transverse roughness gives the lowest minimum film thickness and the longitudinal one gives the highest value. This can be explained by analyzing the distribution of the film thickness in individual simulations.

Fig. 8 shows the distribution of film thickness for the example of a Bi-Gaussian height distribution and compares the results for the transverse and longitudinal roughness orientation. It can be seen in Fig. 8a that for the longitudinal roughness there is a preferential flow in the direction of entrainment which has no resistance against flow, and thus leads to a lower film thickness. However, for the same longitudinal roughness there is a higher resistance against the side flow, see Fig. 8b, which leads to higher minimum thickness (which in this study is located on the sides of the contact). For the transverse roughness, this logic is inverse, since the resistance of the flow in the entrainment direction is higher which leads to a higher central film thickness, but the flow to the sides sees no resistance hence it causes a lower minimum film thickness. For further illustration, the difference in film thickness is shown in Fig. 8c. Here, the film thickness for longitudinal roughness is subtracted from the one for transverse roughness. Resulting, positive values are seen at the center of the contact and at the constriction on the trailing edge of the contact. However, negative values are present at the sides indicating that the transverse roughness yields lower values compared to the longitudinal one.

5.2. Stribeck curves with a virtual mini traction machine point contact

Traction curves, namely fluid, asperity, and total traction coefficients, are shown in Fig. 9 as a function of velocity v . Here, the respective traction coefficients are determined by the ratio between traction force obtained from the simulation results and the normal

force, as given in Table 7. The simulation results show that the overall traction for the point EHL contact, see Fig. 9c, is most influenced by the roughness distribution. The same behavior is observed for the asperity traction alone, shown in Fig. 9b, indicating that this type of traction highly defines the overall friction of the contact. All in all, it can be seen that regarding traction the height distribution plays a more significant role when compared to the roughness orientation. We expect that the change in the significance of minimum and central film thickness, as described in the last Section, see Fig. 8, is the reason for this behavior. This observation may change for line contacts due to the smaller significance of the side flow.

6. Discussions and proposals for a load sharing concept in engineering applications with two roughness parameters

It has been shown that a single roughness metric is not sufficient to be used in an engineering equation for a detailed evaluation of friction. Both effects of distribution and anisotropy must be included in the model to capture the response of the system. The distribution can be addressed with parameters of the Abbott-Firestone curve like S_k or S_{pk} (as in the work of Hansen, Björling and Larsson [15]) or, e.g., with the skewness of the height distribution S_{sk} . For experimental MTM tests reported by Dawczyk et al. [33] and adapted by Taylor and Sherrington [7], all friction measurement data will collapse to one master curve using a classical Lambda-parameter definition of $\lambda = h_c/S_q$, equation in Table 2. For this work, our friction data generated via simulation is shown in Fig. 10. Here, the weight factor X_l is defined as

$$X_l = \frac{F_{total} - F_{fluid}}{F_{asperity} - F_{fluid}}, \quad (4)$$

with F_{total} , F_{fluid} , and $F_{asperity}$ as total, fluid, and asperity traction respectively [33]. They were shown as a function of the velocity in Fig. 9.

In Fig. 10a, the weight factor X_l is shown as a function of the Lambda-parameter given as $\lambda = h_c/S_q$, as in the work by Taylor and Sherrington [7]. The results of the nine generated surfaces is shown for different temperatures, 40 °C, 60 °C, 80 °C, and 100 °C. Since for all surfaces, the single roughness metric influencing the Lambda-parameter, the RMS roughness S_q , is the same, the spread of the curves may be caused by the varying roughness orientations and distributions. As discussed, see Section 5.2, the roughness distribution was found to play the more significant role regarding traction curves. Thus, to account for the roughness distribution we modified the Lambda-parameter to include two roughness metrics. In addition to the ratio between the central film thickness and the RMS roughness, a normalized reduced

Table 8
Improved Yasutomi parameter of Tris(2-ethylhexyl) trimellitate (TOTM), based on [67,68].

A_1 (°C)	A_2 (GPa ⁻¹)	B_1 (GPa ⁻¹)	B_2 (-)	C_1 (-)	C_2 (°C)	T_{80} (°C)	μ_g (Pa s)
275.55	0.3173	14.7493	-0.3125	15.9946	24.8719	-76.9553	1.00×10^{12}

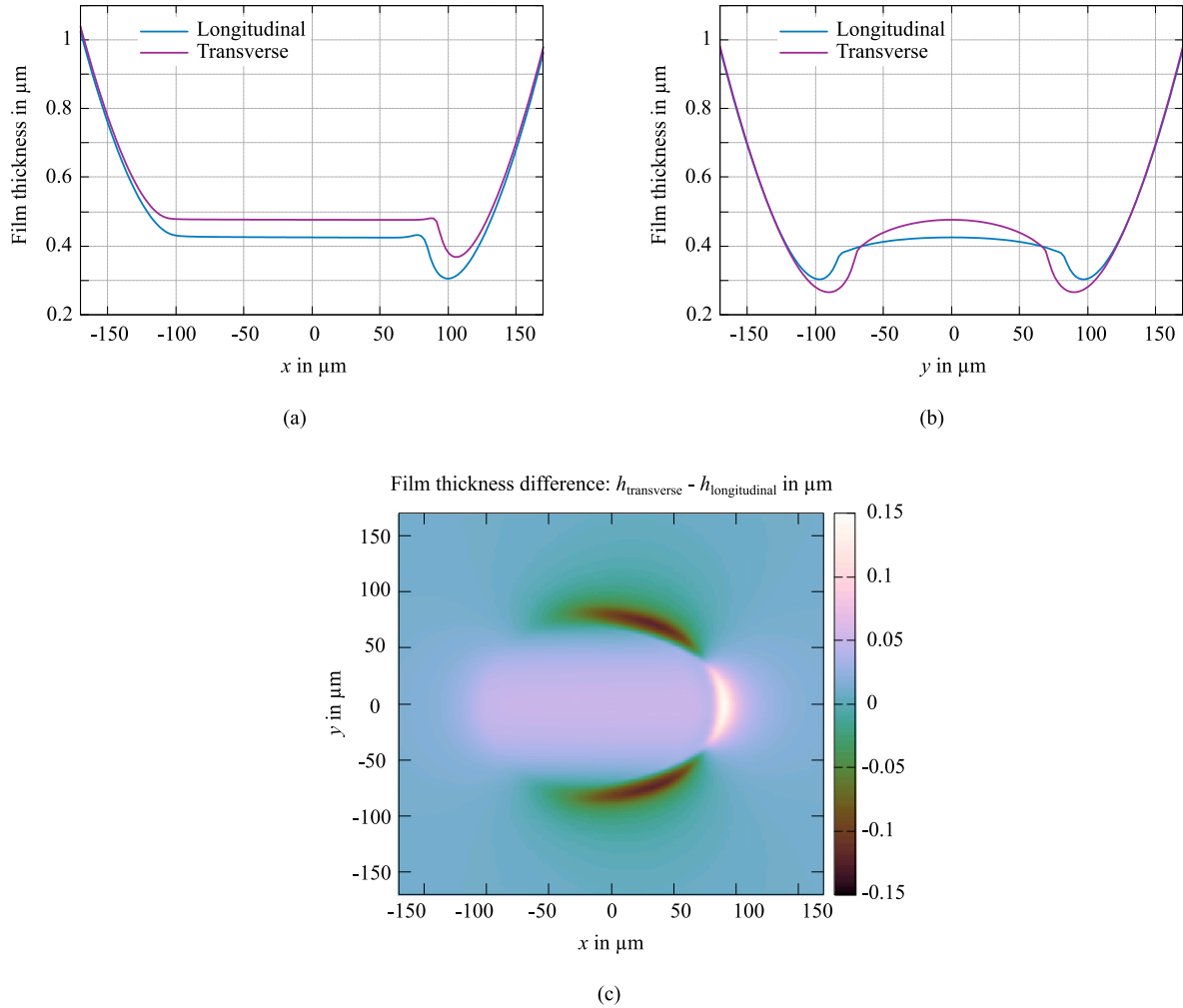


Fig. 8. Distribution of film thickness for comparison of the effect of roughness orientation on different aspects of film thickness in a circular point contact. Results are shown for a Bi-Gaussian surface height distribution, for transverse and longitudinal roughness orientation. (a) In entrainment direction. (b) Perpendicular to entrainment direction. (c) Difference in film thickness, results for longitudinal orientation subtracted from film thickness in the case of transverse orientation.

Table 9
Derived values of viscosity-pressure coefficient α^* and Eyring stress for Tris(2-ethylhexyl) trimellitate (TOTM) based on [67,68].

Temperatures	40 °C	60 °C	80 °C	100 °C
α^* (GPa ⁻¹)	18.2	16.2	14.8	13.5
Eyring stress by $P = 1.0$ GPa (MPa)	5.01	6.60	8.60	11.40

peak height is added to include the effect of the height distribution, so that

$$\lambda_{\text{mod}} = \frac{h_c}{S_q} \left(\frac{S_{pk}}{S_{pk,\text{ref}}} \right)^t. \quad (5)$$

The reference value $S_{pk,\text{ref}}$ must be selected as the maximum of S_{pk} values which have been measured in the set of surfaces under investigation, to give the ratio smaller or equal to unity, $S_{pk,\text{ref}} = \max(S_{pk,i})$. In this work, $S_{pk,\text{ref}}$ is assumed to be $0.089 \mu\text{m}$, see Table 6. Additionally, a fitting parameter t is introduced which we found to be $t = -0.22$ for our load sharing curves. This parameter is specific for this data set and needs to be determined for different applications.

The load sharing curve in Fig. 10c is shown as a function of the modified Lambda-parameter. The spread of the curves is significantly smaller, especially when comparing the results for a single roughness orientation. Exemplary, Fig. 10b and d show the curves for the transverse roughness orientation only, while the height distribution and temperatures are varied. Here, in Fig. 10b the Lambda-parameter is $\lambda = h_c/S_q$, while in Fig. 10d the modified Lambda-parameter is used, as given in Eq. (5).

With the modified Lambda-parameter, the load sharing curves can be approximated with

$$X = \frac{1}{1 + a\lambda_{\text{mod}}^b}. \quad (6)$$

Fitting our results with the equation proposed by Taylor and Sherrington [7] (see Table 2) did not give sufficient matching, which may be caused by the application of the modified Lambda-parameter. However, the above shown equation gives good results while still keeping the number of fitting parameters to two. The fitted curve is shown in Fig. 10d in light color, where the parameters have been fitted using the

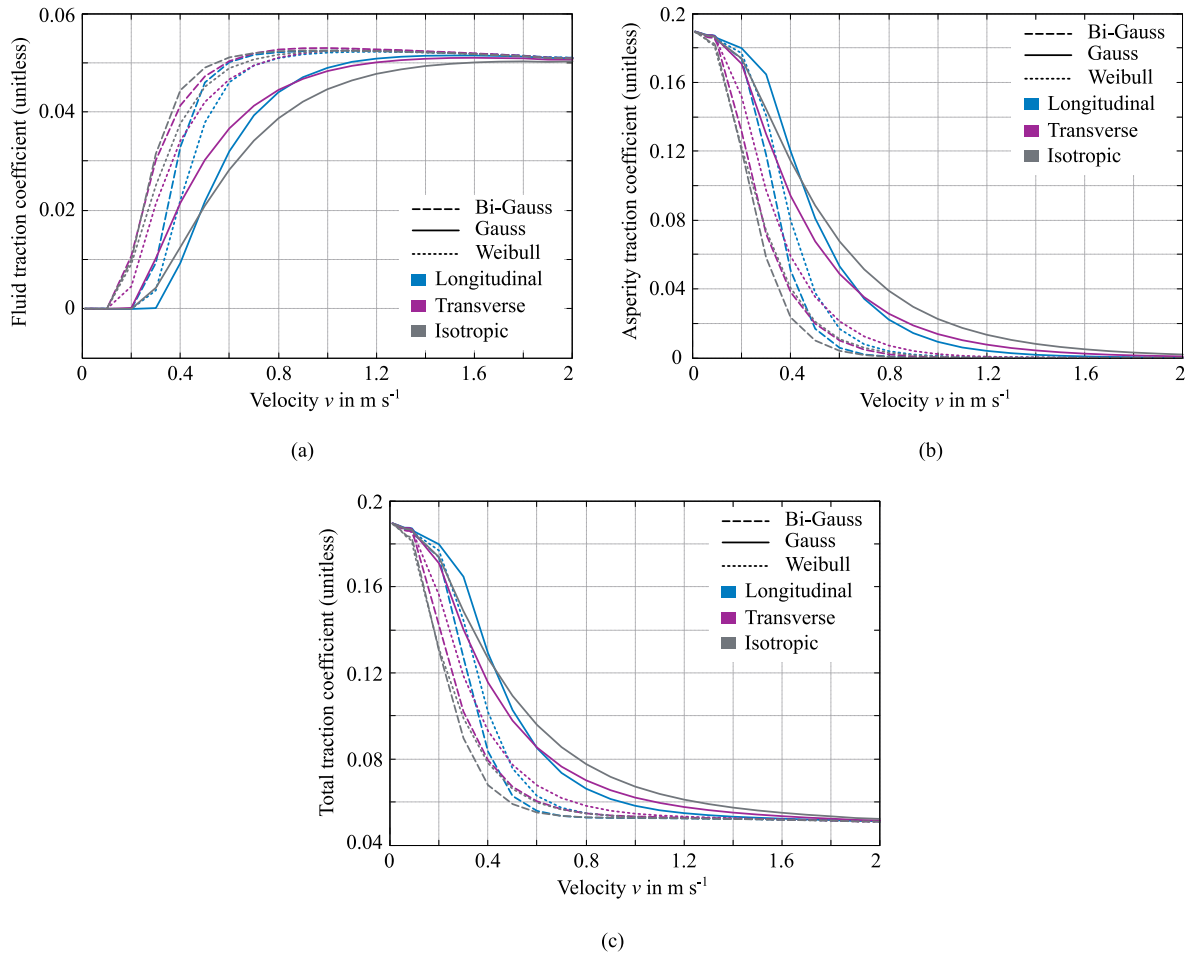


Fig. 9. Traction curves in dependence of velocity v . (a) Fluid traction coefficient. (b) Asperity traction coefficient. (c) Total traction coefficient.

transverse roughness orientation only. Here, we found the parameters to be $a = 0.016$, and $b = 4.551$.

The above proposed modification of the Lambda-parameter can be used as a prototype for mixed friction modeling in which an additional parameter could be used to capture the effect of the height distribution of the roughness. In the framework of this paper we used the simulated traction curves as demonstration to make a proof of concept.

7. Conclusions

In the first part of the paper, we demonstrated why the ratio of film thickness to a single roughness metric (Lambda-parameter) could lead to inconsistent results and does not offer an adequate and sufficient model for load sharing. By the same token, this parameter is not adequate for comparing the roughness patterns (e.g., grinding qualities) concerning mixed-EHL friction. This issue has been addressed from different point of views which are significant in engineering applications. This paper aims to raise this issue in the machine design community to concretely define the type of film thickness and roughness in terms of filtering, measurement method, scale, and status of running-in. It also brings into attention the inconsistencies in the notion of load sharing (load, area, or other sharing concepts) for future research papers in the area of mixed friction. It has also been shown that more than one roughness parameter is needed in order to capture the major characteristics of surfaces in terms of roughness distribution and orientation (anisotropy).

In order to approach the unresolved issues addressed above, artificial surfaces have been generated with the same parameters of power spectral density (included length scales, and same roughness metrics

of RMS roughness S_q , and average roughness S_a) in order to study both effects of roughness distribution as well as anisotropy independent of roughness metrics. The surfaces are uploaded in the online platform <https://contact.engineering/> to be used for other researchers to continue this studies, see Appendix.

Stochastic mixed-EHL simulations have been performed using the average flow model to study the effect of roughness distribution and anisotropy on film thickness and friction. As a model lubricant for simulations, a well characterized and well defined pure substance (TOTM) has been used since the rheological parameters are known. It allows other researchers to run simulations based on the same lubricant, or run experiments and compare the results.

In flow factor calculation, the distribution plays almost no role in pressure flow factors and a rather minor role in shear flow factors. In contrast, for compliance-pressure curves for asperity load sharing the distribution plays the major role and the Weibull and Bi-Gaussian surfaces show a very different pattern compared to the ones with Gaussian distributions. For area sharing again the contrast between effect of distribution and isotropy is not very clear but generally the effect of distribution is more pronounced.

The actual central film thickness after considering the effect of flow factors and averaging the deformed asperities is mainly determined by anisotropy and not by distribution. The longitudinal roughness gives the lowest central film thickness. In contrast, for the minimum film thickness the lowest value is achieved by the transverse roughness distribution. It has been explained by the effect of the resistance to flow in the side direction for the case of point contact. In transversal point contacts, the preferential flow is perpendicular to the entrainment so it gives a smaller film thickness at the sides of the point contact.

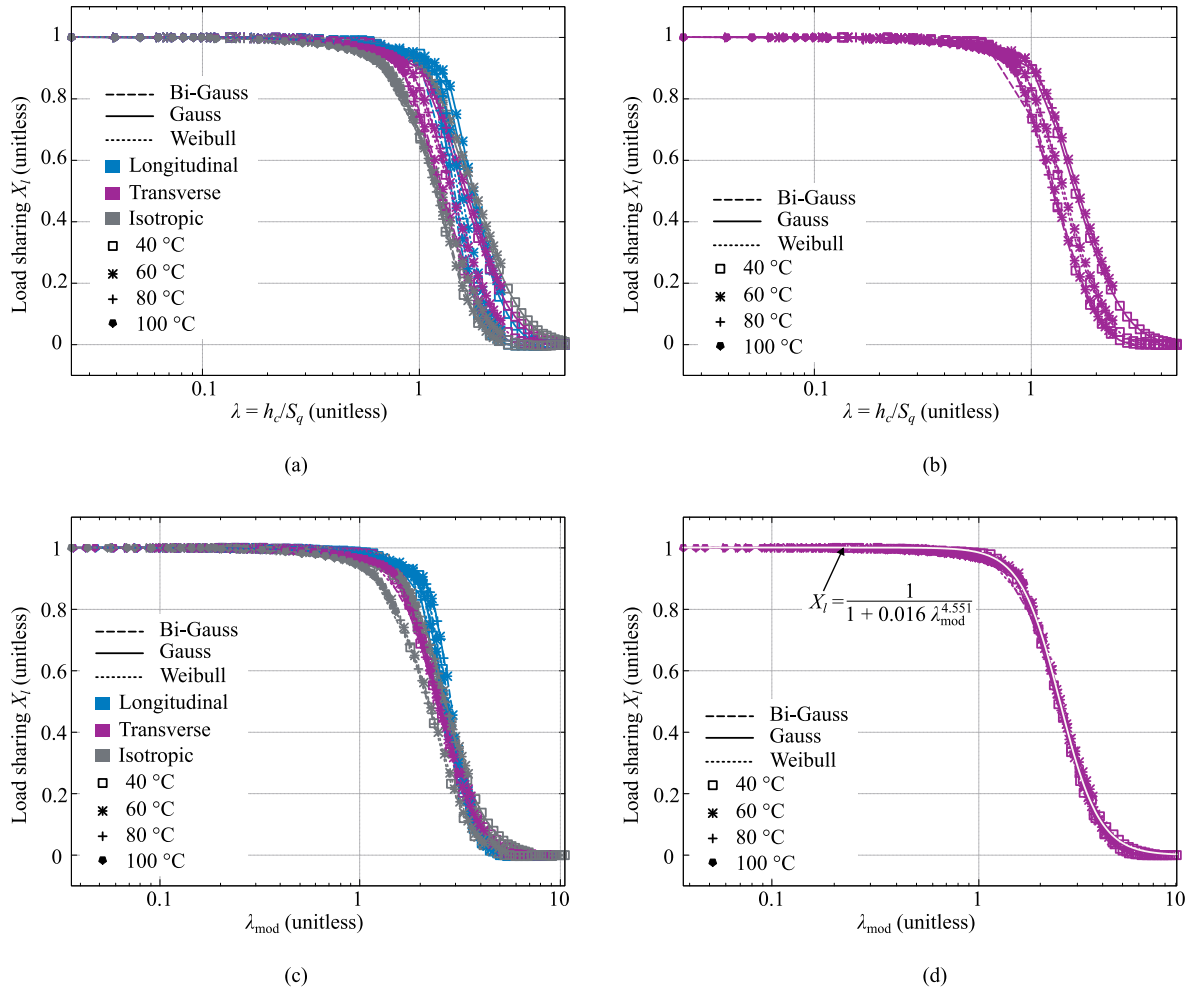


Fig. 10. Load sharing curves for the artificially generated surfaces with varying surface height distribution and roughness orientation, at four different temperatures: 40 °C, 60 °C, 80 °C, and 100 °C, as functions of different definitions of the Lambda-parameter. (a) Classic Lambda-parameter, $\lambda = h_c/S_q$. (b) Classic Lambda-parameter, transverse roughness orientation only. (c) Modified Lambda-parameter, Eq. (5) with $t = -0.22$. (d) Modified Lambda-parameter, Eq. (5) with $t = -0.22$, transverse roughness orientation only, and fitted curve, Eq. (6) with $a = 0.016$ and $b = 4.551$.

The different height distributions (Bi-Gauss, Gauss, Weibull) are distinguished by different line types, and the roughness orientations (longitudinal, transverse, isotropic) are shown by the line color, resulting in the nine artificially generated surfaces. The four different temperatures for each surface are distinguished by symbols, shown in the legend.

The roughness orientations in point contact manifest themselves in an equivocal manner and thus render an objective ranking of the comparison of friction by anisotropy rather difficult. This finding has a significant implication on interpretation of mixed friction results with point contact, e.g., mini traction machine (MTM) tests. In contrast, roughness distribution can be readily recognized as a clear influencing factor. By surface polishing or multiple grinding sequences both the roughness amplitude as well as its distribution cause a significant change in mixed friction.

We have demonstrated that all the simulated friction curves in all temperatures by transverse roughness can be collapsed in one master curve if we consider a modified Lambda-parameter with one additional roughness parameter (in this case a normalized S_{pk}). The additional parameter however could be formulated in terms of any other parameter which is qualified to differentiate between desired futures.

Finally, this paper is a call that every standard or technical specification which uses the ratio of film thickness with roughness, in order to be technically coherent, needs to specify precisely which roughness metrics have to be used and how they should be measured and filtered.

This is also true for the case of film thickness and which method have to be adopted for the particular case.

CRediT authorship contribution statement

Charlotte Spies: Writing – original draft, Visualization, Software, Investigation, Data curation, Conceptualization. **Yash Parab:** Writing – review & editing, Software, Investigation, Data curation. **Arshia Fatemi:** Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table A.10

DOIs of the nine artificially generated surfaces.

Surface	DOI
Gaussian height distribution, Isotropic roughness orientation	10.57703/ce-v2q9g
Gaussian height distribution, Longitudinal roughness orientation	10.57703/ce-wqcvh
Gaussian height distribution, Transverse roughness orientation	10.57703/ce-nbtar
Bi-Gaussian height distribution, Isotropic roughness orientation	10.57703/ce-ydtmz
Bi-Gaussian height distribution, Longitudinal roughness orientation	10.57703/ce-ck7rp
Bi-Gaussian height distribution, Transverse roughness orientation	10.57703/ce-5jyps
Weibull height distribution, Isotropic roughness orientation	10.57703/ce-jhcq8
Weibull height distribution, Longitudinal roughness orientation	10.57703/ce-bgpe9
Weibull height distribution, Transverse roughness orientation	10.57703/ce-rmtpy

Appendix. Access to artificially generated surfaces

The nine artificially generated surfaces are available for download via <https://contact.engineering/>. The characteristics of the surfaces were presented in Section 3. Table A.10 gives the respective DOIs of each surface.

Data availability

Data will be made available on request.

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